

CONTROLLED RESIDUALS IN GAUSSIAN RAMSEY CONSTRUCTIONS

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ABSTRACT. We refine the Gaussian random-graph framework for fixed-ratio Ramsey lower bounds. Relative to the pinned Appendix-B induction of Hunter, Milojević and Sudakov and the exact cumulant input of Lin and Niu, we introduce a nonexchangeable triangular weighted ledger with a deterministic boundary residual. Exact gradient merge uniquely fixes the triangular array and gives only a $(96 \log C)^{-1}$ leading contribution. By retaining the full-box strong-concavity term, we pay a controlled failure of exact merge by square completion and obtain an additional term

$$\hat{H}_*(C) = \frac{1 + o(1)}{64 \log C} > 0.$$

Under the explicit source interface (S1)–(S5), for every sufficiently large fixed C , the resulting source-relative bound is

$$\liminf_{\ell \rightarrow \infty} \frac{1}{\ell} \log R(\ell, \lfloor C\ell \rfloor) \geq -\frac{1}{2} \log p_C + \frac{B_R(C)}{2} + G_*(C) + \hat{H}_*(C).$$

The threshold in C is existential and non-effective. We also identify a narrow scalar residual class for which the coefficient $1/64$ is optimal; the statement does not cover adaptive or higher-order potentials.

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1. INTRODUCTION

The probabilistic argument of Erdős gives the classical exponential scale for diagonal and fixed-ratio Ramsey lower bounds [Erd47]. Ma, Shen and Xie recently obtained the first exponential improvement for $R(\ell, C\ell)$ at every fixed $C > 1$ [MSX25]. Hunter, Milojević and Sudakov gave a Gaussian random-graph construction and stronger quantitative estimates [HMS25]; Lin and Niu sharpened the truncated Gaussian cumulant analysis [LN26]. Truncated Gaussian variance proxies and cumulants also have an independent probabilistic literature [BMA24].

This paper works inside the explicit HMS Appendix-B induction. The new ingredient is not history conditioning by itself—that is already present in the cited constructions—but a deterministic nonexchangeable triangular mean ledger. A boundary derivative must normally match a projection weight exactly. We first show that this exact-merge requirement uniquely determines the array and leaves no scalar optimization freedom. We then retain the negative quadratic term from strong concavity and intentionally permit a small deterministic mismatch. Square completion pays this residual and creates a larger net gain.

1.1. Main theorem. For $C > 1$, let $p_C \in (0, 1/2)$ be the unique solution of

$$C = \frac{\log p_C}{\log(1 - p_C)}.$$

The parameters p, c, D, ω_0 , the source quantities $B_R(C)$ and $B_B(C)$, and the exact-CGF term $G_*(C)$ are defined explicitly in Section 2. The theorem imports the four version-pinned conditional probability, extraction, and blue-ledger items (S1), (S2), (S4), and (S5), together with the separately identified weighted-extension rule (S3), all stated in Source interface 2.2. The one-step nonuniform estimate, first-moment conversion, and red-blue comparison are proved locally.

Theorem 1.1 (Source-relative fixed-ratio bound). *Under Source interface 2.2, there exists $C_0 > 1$ such that, for each fixed $C \geq C_0$,*

$$\liminf_{\ell \rightarrow \infty} \frac{1}{\ell} \log R(\ell, \lfloor C\ell \rfloor) \geq -\frac{1}{2} \log p_C + \frac{B_R(C)}{2} + G_*(C) + \hat{H}_*(C), \quad (1.1)$$

where $\hat{H}_*(C) > 0$ and

$$\hat{H}_*(C) = \frac{1 + o(1)}{64 \log C} \quad (C \rightarrow \infty). \quad (1.2)$$

The public coarsening $B_R(C)/2 \geq 1/20$ may be substituted in (1.1).

The quantifiers are important: C is fixed before $\ell \rightarrow \infty$, and the proof does not give a numerical C_0 . The source contains unspecified $O(\cdot)$ and $o(\cdot)$ constants, so an effective threshold cannot be extracted from the pinned text alone.

1.2. Contributions.

- (i) We prove rigidity of the exact-gradient triangular recursion.
- (ii) We construct an adapted deterministic residual and prove a uniform full-box Hessian bound.
- (iii) We compute the exact multiplicity of the residual and its square penalty.
- (iv) We insert a three-factor Hölder decomposition without counting the old exact-CGF gain twice.
- (v) We propagate the ledger through reverse induction and quantify the perfect-sequence extraction loss.
- (vi) We optimize a narrowly specified scalar residual class and obtain the leading coefficient $1/64$.

The result is stated as a source-relative strengthening. A direct claim of improvement over every printed final normalization in Lin–Niu would require both expressions to be derived from one common terminal probability formula; that comparison is not asserted here.

2. GAUSSIAN SOURCE LEDGER AND NOTATION

Fix $C > 1$. Let $p_C \in (0, 1/2)$ be the solution of

$$\log p_C = C \log(1 - p_C),$$

and, in the range of C used below, set

$$p = p_C + \frac{1}{C} < \frac{1}{2}, \quad \Phi(-c) = p, \quad a = \varphi(c), \quad D = \frac{4aC}{1-p}. \quad (2.1)$$

For the fixed- C limit $\ell \rightarrow \infty$, take

$$d = \lceil D^2 \ell^2 \rceil, \quad D_\ell = \frac{\sqrt{d}}{\ell} = D + O_C(\ell^{-2}). \quad (2.2)$$

The harmless replacement of D by $D_\ell \geq D$ in finite- ℓ feasibility inequalities will always be left implicit.

For a standard normal variable conditioned on $Z \leq -b$, define

$$m(b) = \frac{\varphi(b)}{\Phi(-b)}, \quad v(b) = 1 + b m(b) - m(b)^2, \quad Y_b = Z + m(b), \quad (2.3)$$

so that Y_b is centered and has variance $v(b)$. Its positive-direction cumulant generating function is

$$K_b(u) = \log \mathbb{E} e^{u Y_b} = \frac{u^2}{2} + u m(b) + \log \Phi(-b - u) - \log \Phi(-b), \quad u \geq 0, \quad (2.4)$$

and we write

$$\psi_b(u) = \begin{cases} K_b(u)/u^2, & u > 0, \\ v(b)/2, & u = 0. \end{cases} \quad (2.5)$$

We shall repeatedly use

$$K_b''(u) = v(b + u), \quad K_b(u) = \int_0^u (u - s) v(b + s) ds. \quad (2.6)$$

Lemma 2.1 (One-sided Gaussian calculus). *For every $b \in \mathbb{R}$, $m'(b) = m(b)(m(b) - b) > 0$, $m''(b) \geq 0$, and $v(b) = 1 - m'(b)$ is nonincreasing. Consequently, $0 < v(b) \leq 1$, $K_b''(u) \in [0, 1]$ for $u \geq 0$, and $\psi_b(u)$ is nonincreasing in both b and u .*

Proof. Differentiating $m = \varphi/\Phi(-\cdot)$ gives $m' = m(m - b)$. The classical inverse-Mills inequality $m(b) > b$ for $b > 0$ (and positivity of m for $b \leq 0$) gives $m' > 0$. Moreover $v = 1 - m'$. Monotonicity of the variance of an upper-truncated normal in its upper cutoff, applied at the cutoff $-b$, says that v is nonincreasing in b ; equivalently $m'' = -v' \geq 0$. The variance bounds give $0 < v \leq 1$. Formula (2.6) then implies the assertions concerning K_b and ψ_b ; for example $\psi_b(u) = \int_0^1 (1 - t) v(b + ut) dt$. $\square$

Put

$$\begin{aligned} m_0 &= m(c) = \frac{a}{p}, & m'_0 &= m'(c), \\ \alpha_R &= 10\sqrt{\log(10/p)}, & \omega_0 &= \frac{\alpha_R^2}{D} = \frac{100\log(10/p)}{D}. \end{aligned} \quad (2.7)$$

and

$$\rho = \frac{m_0 m'_0}{D}, \quad \theta = m_0^2, \quad A_0 = m_0^2 m'_0. \quad (2.8)$$

For fixed $w > \omega_0$, define the lower and upper cutoff extrema

$$\begin{aligned} m_w &= m(c - w), & M_w &= m(c + w), \\ j_-(w) &= m'(c - w), & j_+(w) &= m'(c + w). \end{aligned} \quad (2.9)$$

The distinction between m_w and M_w is essential: the former gives the history-uniform exact-CGF deficit, whereas the latter bounds the new linear tilts.

2.1. The frozen red and blue source ledgers. Fix one absolute source constant K that dominates the red $O(\sqrt{\log(1/p)}/D)$ errors in the pinned HMS induction, and put

$$\vartheta_C = 1 - K \frac{\sqrt{\log(1/p)}}{D} > 0, \quad (2.10)$$

$$B_R(C) = -\log \frac{p}{p_C} + \vartheta_C \frac{a^3}{3p^3 D}, \quad B_B(C) = \log \frac{1 - p_C}{1 - p} - \frac{4a^3 C}{3(1 - p)^3 D}. \quad (2.11)$$

The literal red target envelope is

$$\mathcal{H}_C(r) = p^{(r)} \exp \left[-\vartheta_C \frac{a^3}{p^3 \sqrt{d}} \binom{r}{3} + \frac{a}{p} d^{-1/5} \binom{r}{2} + r \right]. \quad (2.12)$$

Thus $B_R/2$ and $CB_B/2$, respectively, are the exact limiting red and blue source bonuses relative to $-\frac{1}{2} \log p_C$. In particular, these symbols do not denote the conclusion of the present paper.

2.2. The pre-existing exact-CGF deficit. For $1 \leq k \leq \ell$, set

$$P_{k,d}^0 = \frac{\sqrt{d}}{\sqrt{d} - 4m_0 k}, \quad (2.13)$$

and, whenever the denominator is positive, define

$$\mathbf{d}_{k,w,d} = \frac{m_0^2 m_w^2}{d} k(k-1)^2 \left[1 - P_{k,d}^0 \psi_{c-w} \left(\frac{P_{k,d}^0 m_0 m_w (k-1)}{\sqrt{d}} \right) \right], \quad (2.14)$$

$$\mathcal{D}_{C,w,d}(r) = \sum_{k=1}^{r-1} \mathbf{d}_{k,w,d}. \quad (2.15)$$

For $0 \leq x \leq 1$, let

$$P_*(x) = \frac{D}{D - 4m_0 x}, \quad u_*(x) = \frac{P_*(x) m_0 m(c - \omega_0) x}{D}. \quad (2.16)$$

For all sufficiently large C , $D > 8m_0$ and $P_*(1)v(c - \omega_0) < 2$; hence (2.13)–(2.17) have positive denominators and nonnegative integrands throughout the range used below. The exact source-relative CGF term used in the theorem is

$$G_*(C) = \frac{m_0^2 m (c - \omega_0)^2}{D^2} \int_0^1 x^3 [1 - P_*(x) \psi_{c-\omega_0}(u_*(x))] dx. \quad (2.17)$$

Equivalently, if $G(C, w)$ is obtained from (2.17) by replacing ω_0 with w , then

$$\lim_{\ell \rightarrow \infty} \ell^{-2} \mathcal{D}_{C,w,d}(\ell) = G(C, w), \quad G_*(C) = \lim_{w \downarrow \omega_0} G(C, w). \quad (2.18)$$

2.3. The controlled-residual constants. Set

$$a_- = \min\{A_0, \theta\}, \quad a_+ = \max\left\{\theta, \frac{A_0 + \rho\theta}{1 - \rho}\right\}, \quad (2.19)$$

$$\mu_w = j_-(w) - j_+(w)^2 \frac{m_0 + \theta/D}{D(1 - \rho)}, \quad (2.20)$$

and

$$\hat{L}_w(C) = \frac{4m_0 M_w^2}{D} + \frac{2M_w^2 a_+}{D^2} + \frac{32m_0^2 M_w^2 a_+}{a_- D^2}. \quad (2.21)$$

Finally put

$$\hat{B}_w(C) = m_0^2 - \frac{16}{3} - \hat{L}_w(C), \quad E_w(C) = \hat{B}_w(C)\theta - \frac{\theta^2}{2\mu_w}, \quad (2.22)$$

$$\hat{H}(C, w) = \frac{\hat{B}_w(C)A_0 + E_w(C)}{12D^2}, \quad \hat{H}_*(C) = \lim_{w \downarrow \omega_0} \hat{H}(C, w). \quad (2.23)$$

2.4. Probability space, exposure histories, and source events. For each $0 \leq r \leq \ell$, let $\mathbf{x}_1, \dots, \mathbf{x}_r$ be independent $N(0, d^{-1}I_d)$ vectors. An edge is blue when $\langle \mathbf{x}_i, \mathbf{x}_j \rangle \geq -c/\sqrt{d}$ and red otherwise. We use the Bartlett representation $\mathbf{y}_1, \dots, \mathbf{y}_r$: the entries $y_j(i)$, $i < j$, are independent $N(0, d^{-1})$, the diagonal entries are independent with $y_i(i) \sim \sqrt{\chi_{d-i+1}^2/d}$, and $y_j(i) = 0$ for $i > j$. This representation has the same joint Gram matrix as the original Gaussian vectors.

Let $M_i = (y_j(i))_{j=i+1}^r$ be the strict part of column i , and put

$$\mathcal{D}_r = \sigma(y_1(1), \dots, y_r(r)).$$

Define the exposure filtration

$$\mathcal{F}_s = \mathcal{D}_r \vee \sigma(M_1, \dots, M_s), \quad 0 \leq s \leq r. \quad (2.24)$$

All conditional assertions below are statements about regular conditional probabilities and hold almost surely. Write π_s for projection onto the first s Bartlett coordinates and put

$$h_{ij} = \langle \pi_{i-1}(\mathbf{y}_i), \pi_{i-1}(\mathbf{y}_j) \rangle, \quad 1 \leq i < j \leq r. \quad (2.25)$$

Thus h_{ij} is $\mathcal{F}_{i-1}$ -measurable. Let E_{ij} be the blue-edge event and set

$$I_s = \bigcap_{s < i < j \leq r} \bar{E}_{ij}, \quad C_s = \bigcap_{s < i < j \leq r} E_{ij}. \quad (2.26)$$

For the source perfectness events put $\alpha_B = 10\sqrt{C \log(10/p)}$ and $\delta = \alpha_B d^{-1/4}$. The event B_r^R says that, for every $i \leq r$,

$$\|\mathbf{y}_i\| \in (1 - \delta, 1 + \delta), \quad \|\pi_{i-1}(\mathbf{y}_i)\| \leq \frac{\alpha_R \sqrt{\ell}}{\sqrt{d}}; \quad (2.27)$$

B_r^B is defined by replacing α_R with α_B . We use the probability notation

$$P_{R,r}^* = \mathbb{P}(I_0 \cap B_r^R), \quad P_{R,r} = \mathbb{P}(I_0), \quad P_{B,r}^* = \mathbb{P}(C_0 \cap B_r^B), \quad P_{B,r} = \mathbb{P}(C_0). \quad (2.28)$$

Empty intersections have probability one.

For $0 \leq s \leq r$, let

$$\kappa_{r,s}^R = \mathbb{P}(B_r^R \mid \mathcal{F}_s), \quad \mathfrak{H}_{r,s}^R = \{\kappa_{r,s}^R > 0\}. \quad (2.29)$$

We call an $\mathcal{F}_s$ -history *red-admissible* when it belongs to $\mathfrak{H}_{r,s}^R$. This definition is independent of a choice of coordinates for the conditional kernel. On the complementary histories, $\mathbb{P}(I_s \cap B_r^R \mid \mathcal{F}_s) = 0$ almost surely. Thus every conditional estimate below only needs to be checked on $\mathfrak{H}_{r,s}^R$.

At exposure i , let $A_i^R = \bigcap_{j>i} \bar{E}_{ij}$. Conditional on A_i^R and $\mathcal{F}_{i-1}$, set $X_j = y_j(i)$, $j > i$. Its exact upper cutoff is

$$X_j \leq -\frac{b_{ij}}{\sqrt{d}}, \quad b_{ij} = \frac{c + \sqrt{d} h_{ij}}{y_i(i)}. \quad (2.30)$$

The red-perfect extraction map Γ_R scans $\mathbf{x}_1, \dots, \mathbf{x}_\ell$ in order and retains an index when adjoining its vector preserves the two inequalities in (2.27). For a specified $I \subseteq [\ell]$, define

$$p_I = \mathbb{P}(I_0 \cap \{\Gamma_R = I\}). \quad (2.31)$$

This fixes both the event and the retained-order relabeling used later.

For a deterministic nonnegative triangular array $T = (T_{jq})_{1 \leq j < q \leq r}$, define the weighted projection potential

$$\mathcal{P}_s^T = \sum_{s < j < q \leq r} T_{jq} \langle \pi_s(\mathbf{y}_j), \pi_s(\mathbf{y}_q) \rangle. \quad (2.32)$$

For $i \geq 1$, its normalized one-step kernel is

$$\begin{aligned} \mathcal{K}_i^T &= \exp\left(\sum_{j>i} T_{ij} h_{ij}\right) \mathbb{P}(A_i^R \mid \mathcal{F}_{i-1}) \\ &\times \mathbb{E}\left[\exp\left\{-\sum_{i < j < q \leq r} T_{jq} X_j X_q\right\} \middle| A_i^R, \mathcal{F}_{i-1}\right]. \end{aligned} \quad (2.33)$$

Source interface 2.2 (Version-pinned source and weighted-extension interface). Items (S1), (S2), (S4), and (S5) are imported from the content-addressed HMS v2 and Lin–Niu v2 sources identified in Appendix B. Item (S3) is the exact weighted-extension hypothesis required by this paper; it is not represented as a theorem stated verbatim in either source.

- (S1) For every $r \leq \ell$ and $1 \leq i < r$, on every red-admissible $\mathcal{F}_{i-1}$ -history, the next-column coordinates in (2.30), conditionally on A_i^R , are independent $N(0, d^{-1})$ variables truncated by the displayed upper cutoffs, and

$$|b_{ij} - c - \sqrt{d} h_{ij}| \leq d^{-1/5}, \quad |\sqrt{d} h_{ij}| \leq \omega_0. \quad (2.34)$$

Thus, for every fixed $w > \omega_0$, all cutoffs lie in $[c - w, c + w]$ once $\ell \geq \ell_0(C, w)$.

- (S2) Under the conditioning and history quantifiers in (S1), with $\zeta_j = \sqrt{d}(X_j - \mathbb{E}[X_j \mid A_i^R, \mathcal{F}_{i-1}])$, the variables $(\zeta_j)_{j>i}$ are conditionally independent, centered, and 1-subgaussian, and

$$\log \mathbb{E}[e^{x\zeta_j^2} \mid A_i^R, \mathcal{F}_{i-1}] \leq \frac{16}{3}x, \quad 0 \leq x \leq \frac{1}{8}. \quad (2.35)$$

For $k = r - i$, $Q_0 = d/(4t_0k)$, and centered ζ_j , the frozen uniform-quadratic factor obeys

$$\frac{1}{Q_0} \log \mathbb{E} \exp \left\{ -\frac{Q_0 t_0}{d} \sum_{i < j < q \leq r} \zeta_j \zeta_q \right\} \leq \frac{4t_0k}{d}. \quad (2.36)$$

The positive-direction CGF is exactly (2.4). In the source pre-Cauchy allocation, its centered-linear proxy is precisely $\sum_{j>i} (u_j^0)^2$, with u_j^0 as in (5.9); replacing that proxy by the exact CGF produces the same-history deficit (5.16).

- (S3) For every target $r \leq \ell$, every deterministic nonnegative triangular T fixed before exposure, and every deterministic real sequence $\lambda_1, \dots, \lambda_{r-1}$, the following weighted reverse-induction rule holds. If, on every history in $\mathfrak{H}_{r,i-1}^R$, with $k = r - i$,

$$\mathcal{K}_i^T \leq \frac{\mathcal{H}_C(k+1)}{\mathcal{H}_C(k)} e^{-\lambda_i}, \quad 1 \leq i < r, \quad (2.37)$$

then, for every $0 \leq s \leq r$,

$$\mathbb{P}(I_s \cap B_r^R \mid \mathcal{F}_s) \leq \mathcal{H}_C(r-s) \exp \left[-\mathcal{P}_s^T - \sum_{i=s+1}^{r-1} \lambda_i \right]. \quad (2.38)$$

On $(\mathfrak{H}_{r,s}^R)^c$, the left side of (2.38) is zero, so the same inequality holds almost surely on the entire conditional state space. The source law of total probability is

$$\mathbb{P}(I_{i-1} \cap B_r^R \mid \mathcal{F}_{i-1}) = \mathbb{P}(A_i^R \mid \mathcal{F}_{i-1}) \mathbb{E}[\mathbb{P}(I_i \cap B_r^R \mid \mathcal{F}_i) \mid A_i^R, \mathcal{F}_{i-1}], \quad (2.39)$$

and the exact projection identities are

$$\begin{aligned}\mathcal{P}_i^T &= \sum_{i < j < q \leq r} T_{jq} \langle \pi_{i-1} \mathbf{y}_j, \pi_{i-1} \mathbf{y}_q \rangle + \sum_{i < j < q \leq r} T_{jq} X_j X_q, \\ \mathcal{P}_{i-1}^T &= \sum_{i < j < q \leq r} T_{jq} \langle \pi_{i-1} \mathbf{y}_j, \pi_{i-1} \mathbf{y}_q \rangle + \sum_{j > i} T_{ij} h_{ij}.\end{aligned}\tag{2.40}$$

Equations (2.33), (2.39), and (2.40) prove (2.38) by reverse induction from $s = r$. For the uniform array $T_{jq}^0 = t_0$, the version-pinned source scalar allocation, together with the exact-CGF replacement in (S2), is the explicit baseline statement on every history in $\mathfrak{H}_{r,i-1}^R$:

$$\mathcal{K}_i^{T^0} \leq \frac{\mathcal{H}_C(k+1)}{\mathcal{H}_C(k)} e^{-\mathbf{d}_{k,w,d}} \quad (k = r - i).\tag{2.41}$$

Equation (2.41), rather than a final Ramsey bound, is the scalar source input. No nonuniform one-step estimate is imported. Sections 4–6 compare the nonuniform kernel with (2.41) and prove (2.37) for the array used here.

- (S4) If red-perfect extraction deletes u vertices from a target of size ℓ , then, for each specified retained set I of size $\ell - u$,

$$p_I \leq P_{R,\ell-u}^* \left(\frac{p}{10} \right)^{10\ell u},\tag{2.42}$$

for every $0 \leq u \leq \ell$, with $P_{R,0}^* = \mathcal{H}_C(0) = 1$. The probability $P_{R,\ell-u}^*$ uses the retained-order relabeling defined after (2.31).

- (S5) For each fixed C , there is a real sequence $\epsilon_{C,\ell} \rightarrow 0$ such that, with $q_\ell = \lfloor C\ell \rfloor$, the unchanged blue induction and its perfect-sequence extraction give

$$P_{B,q_\ell} \leq (1-p)^{\binom{q_\ell}{2}} \exp \left[\frac{4a^3}{(1-p)^3 \sqrt{d}} \binom{q_\ell}{3} + \epsilon_{C,\ell} \ell^2 \right].\tag{2.43}$$

No Ramsey-rate conclusion, no red–blue comparison, and no new residual term is included in (S1), (S2), (S4), or (S5). The only additional source-side hypothesis is the precisely delimited weighted rule (S3).

The constant K in (2.10) is not effective in the pinned source. Consequently all sufficiently-large- C statements below are existential.

3. RIGIDITY OF EXACT GRADIENT MERGE

For a target ordered set $[r]$, consider deterministic triangular weights W_{ij} and the connection-plus-mean potential

$$F_i(b) = \sum_{j > i} \log \Phi(-b_j) - \frac{1}{d} \sum_{i < j < q} W_{jq} m(b_j) m(b_q).\tag{3.1}$$

At the central cutoff,

$$\partial_j F_i(c\mathbf{1}) = -m_0 - \frac{m_0 m'_0}{d} \sum_{\substack{q>i \\ q \neq j}} W_{jq}. \quad (3.2)$$

Proposition 3.1 (Unique exact-merge array). *Normalize the coefficient of $\sum_{j>i} \log \Phi(-b_j)$ in (3.1) to be one. If the linear term in (3.2), multiplied by $b_j - c = \sqrt{d} h_{ij}$, must equal $-W_{ij} h_{ij}$ at every boundary, then the weights are uniquely determined by*

$$W_{ij} = m_0 \sqrt{d} + \frac{m_0 m'_0}{\sqrt{d}} \sum_{\substack{q>i \\ q \neq j}} W_{jq}. \quad (3.3)$$

There is no further normalized scalar degree of freedom.

Proof. The coefficient comparison gives (3.3). Process the first endpoint in decreasing order. Every weight on the right side has already been defined, so the system is triangular and has a unique solution. $\square$

Remark 3.2 (Global presentation scale). Multiplying the entire inequality by $\kappa > 0$ is a presentation equivalence, not a new normalized array. After dividing the connection term back to coefficient one, the normalized array still satisfies (3.3). If one instead calls $W^{\text{eff}} = \kappa W$ the effective coefficient without renormalizing, then its affine recursion has base term $\kappa m_0 \sqrt{d}$, not $m_0 \sqrt{d}$. Thus a normalized solution and a κ -scaled effective coefficient are never identified. No assertion is made about edge-dependent rescalings.

The feedback operator in (3.3) has row norm at most $\rho = m_0 m'_0 / D = o(1)$. Its accumulated increment is

$$2A_0 \binom{r}{4} [1 + O(\rho)]. \quad (3.4)$$

With the source centered coefficient $m_0^2 [1 + o(1)]$, the exact-merge ledger contributes the previously reviewed term $(1 + o(1)) / (96 \log C)$. Exact feedback evaluation can change only the next order.

4. CONTROLLED RESIDUAL AND STRONG CONCAVITY

Set

$$t_0 = m_0 \sqrt{d}, \quad \gamma_d = \frac{m_0 m'_0}{\sqrt{d}}, \quad e_{ij} = \theta(i - 1). \quad (4.1)$$

In decreasing order of the first endpoint define

$$T_{ij} = t_0 + \gamma_d \sum_{\substack{q>i \\ q \neq j}} T_{jq} + e_{ij}. \quad (4.2)$$

The array is deterministic and adapted: when column i is exposed, every future weight in (4.2) has already been fixed.

With F_i as in (3.1), now using T , we obtain the exact residual identity

$$\partial_j F_i(c\mathbf{1}) = -\frac{T_{ij} - e_{ij}}{\sqrt{d}}. \quad (4.3)$$

Thus the mismatch is explicit and deterministic.

Lemma 4.1 (Uniform Hessian bound). *For all sufficiently large fixed C , every $b \in [c - w, c + w]^{V_i}$ satisfies*

$$-\nabla^2 F_i(b) \succeq \mu_w I, \quad (4.4)$$

where $\mu_w > 0$ is defined in (2.20).

Proof. The recursion gives

$$T_{\max} \leq \frac{t_0 + \theta r}{1 - \rho}, \quad \frac{1}{d} \max_j \sum_{q \neq j} T_{jq} \leq \frac{m_0 + \theta/D}{D(1 - \rho)}.$$

The diagonal of $-\nabla^2 F_i$ is at least $j_-(w)$. The absolute off-diagonal row sum is at most the second term subtracted in (2.20), multiplied by $j_+(w)^2$. Gershgorin's theorem proves (4.4). $\square$

Write

$$\delta_j = b_j - c = \sqrt{d} h_{ij} + \varepsilon_{ij}, \quad |\varepsilon_{ij}| \leq d^{-1/5}.$$

Strong concavity, (4.3), and

$$\frac{e_{ij} \delta_j}{\sqrt{d}} - \frac{\mu_w \delta_j^2}{2} \leq \frac{e_{ij}^2}{2\mu_w d} \quad (4.5)$$

give

$$F_i(b) \leq F_i(c\mathbf{1}) - \sum_{j>i} T_{ij} h_{ij} + \sum_{j>i} \frac{e_{ij}^2}{2\mu_w d} + \frac{d^{-1/5}}{\sqrt{d}} \sum_{j>i} T_{ij}. \quad (4.6)$$

The full boundary coefficient merges into the projection potential, while the failure of exact gradient merge is paid by the square term.

5. DETERMINISTIC MASS AND THE CENTERED-MOMENT LEDGER

Write $\Delta_{ij} = T_{ij} - t_0$. The recursion implies

$$A_0(r - j - 1) + \theta(j - 1) \leq \Delta_{jq} \leq \frac{A_0 + \rho\theta}{1 - \rho}(r - j - 1) + \theta(j - 1). \quad (5.1)$$

Consequently

$$a_-(r - 2) \leq \Delta_{jq} \leq a_+(r - 2). \quad (5.2)$$

For $V_i = \{i + 1, \dots, r\}$, of size $k = r - i$, put

$$S_i = \sum_{\{j,q\} \subset V_i} \Delta_{jq}, \quad R_i = \max_{j \in V_i} \sum_{q \neq j} \Delta_{jq}.$$

Then

$$S_i \geq a_-(r - 2) \binom{k}{2}, \quad R_i \leq a_+(r - 2)(k - 1), \quad \frac{R_i}{S_i} \leq \frac{2a_+}{a_-k}. \quad (5.3)$$

Lemma 5.1 (Exact accumulated multiplicities). *Let*

$$K_r = \sum_{i=1}^{r-1} (r-i)(i-1)^2 = \binom{r}{3} + 2\binom{r}{4}. \quad (5.4)$$

Then

$$\sum_{i=1}^{r-1} S_i \geq 2A_0 \binom{r}{4} + \theta K_r, \quad \sum_{1 \leq i < j \leq r} e_{ij}^2 = \theta^2 K_r. \quad (5.5)$$

Proof. Sum the two terms of the lower bound in (5.1) with their exposure multiplicities. The future-degree term counts every ordered quadruple twice, giving $2\binom{r}{4}$; the residual term is exactly the sum in (5.4). The second identity follows directly from $e_{ij} = \theta(i-1)$. $\square$

5.1. Exact conditional factorization. Fix a history in B_r^R at exposure step i , conditionally on A_i^R and $\mathcal{F}_{i-1}$. For $j \in V_i$, let X_j be the coordinate in (2.30), write $b_j = b_{ij}$, put $m_j = m(b_j)$, and standardize its centered part by

$$\zeta_j = \sqrt{d}(X_j - \mathbb{E}X_j) = \sqrt{d}X_j + m_j. \quad (5.6)$$

By (S2), the ζ_j are conditionally independent, centered and 1-subgaussian. Expanding the weighted quadratic form gives the exact identity

$$\begin{aligned} \log \mathbb{E} \exp \left\{ - \sum_{\{j,q\} \subset V_i} T_{jq} X_j X_q \right\} \\ = -\frac{1}{d} \sum_{\{j,q\} \subset V_i} T_{jq} m_j m_q + \Lambda_i, \end{aligned} \quad (5.7)$$

$$\Lambda_i = \log \mathbb{E} \exp \left\{ \begin{aligned} & \sum_{j \in V_i} (u_j^0 + \delta u_j) \zeta_j \\ & - \frac{t_0}{d} \sum_{j < q} \zeta_j \zeta_q - \frac{1}{d} \sum_{j < q} \Delta_{jq} \zeta_j \zeta_q \end{aligned} \right\}. \quad (5.8)$$

where

$$u_j^0 = \frac{t_0}{d} \sum_{q \neq j} m_q = \frac{m_0}{\sqrt{d}} \sum_{q \neq j} m_q, \quad \delta u_j = \frac{1}{d} \sum_{q \neq j} \Delta_{jq} m_q. \quad (5.9)$$

All tilts in (5.9) are nonnegative.

5.2. Three-factor Hölder and the square-MGF cost. For $k \geq 2$, choose

$$Q_0 = \frac{d}{4t_0k}, \quad Q_1 = \frac{d}{4R_i}, \quad P^{-1} = 1 - Q_0^{-1} - Q_1^{-1}. \quad (5.10)$$

If $R_i = 0$, the increment factor is absent and we set $Q_1 = \infty$. For all sufficiently large C ,

$$Q_0^{-1} + Q_1^{-1} \leq \frac{4m_0}{D} + \frac{4a_+}{D^2} < \frac{1}{2}, \quad (5.11)$$

so $Q_0, Q_1 > 1$ and $1 < P \leq 2$. Hölder's inequality applied to the three random factors in (5.8) yields

$$\begin{aligned} \Lambda_i &\leq \frac{1}{P} \sum_j K_{b_j}(P(u_j^0 + \delta u_j)) \\ &\quad + \frac{1}{Q_0} \log \mathbb{E} \exp \left\{ -\frac{Q_0 t_0}{d} \sum_{j < q} \zeta_j \zeta_q \right\} \\ &\quad + \frac{1}{Q_1} \log \mathbb{E} \exp \left\{ -\frac{Q_1}{d} \sum_{j < q} \Delta_{jq} \zeta_j \zeta_q \right\}. \end{aligned} \quad (5.12)$$

The middle line is the unchanged HMS uniform-quadratic factor and costs at most $4t_0 k/d$.

For the last line, write $R_j = \sum_{q \neq j} \Delta_{jq}$. Nonnegativity of Δ gives

$$-\sum_{j < q} \Delta_{jq} \zeta_j \zeta_q \leq \frac{1}{2} \sum_j R_j \zeta_j^2. \quad (5.13)$$

Because $Q_1 R_j / (2d) \leq 1/8$, the pinned square-MGF estimate

$$\log \mathbb{E} e^{x \zeta_j^2} \leq \frac{16}{3} x, \quad 0 \leq x \leq \frac{1}{8}, \quad (5.14)$$

and conditional independence imply

$$\begin{aligned} \frac{1}{Q_1} \log \mathbb{E} \exp \left\{ -\frac{Q_1}{d} \sum_{j < q} \Delta_{jq} \zeta_j \zeta_q \right\} &\leq \frac{1}{Q_1} \sum_j \frac{16}{3} \frac{Q_1 R_j}{2d} \\ &= \frac{16}{3} \frac{S_i}{d}, \end{aligned} \quad (5.15)$$

where $\sum_j R_j = 2S_i$. This derives the constant $16/3$.

5.3. The same-history CGF bridge. Let $P^0 = P_{k,d}^0$, so that $(P^0)^{-1} = 1 - Q_0^{-1}$. The old optimized uniform ledger on this same conditional history has deficit

$$\mathfrak{d}_i(b) = \sum_{j \in V_i} \left[(u_j^0)^2 - \frac{1}{P^0} K_{b_j}(P^0 u_j^0) \right]. \quad (5.16)$$

Equations (2.6), the cutoff window, and $\sum_{q \neq j} m_q \geq (k-1)m_w$ give the deterministic lower bound

$$\mathfrak{d}_i(b) \geq \mathfrak{d}_{k,w,d}. \quad (5.17)$$

Indeed, after writing $K_b(P^0 u)/P^0 = P^0 u^2 \psi_b(P^0 u)$, this is exactly the monotonicity comparison used in (2.14). Thus the term whose accumulated limit is $G_*(C)$ is identified before any new factor is introduced.

It remains to compare the new linear factor with the second term in (5.16). Since $K_b'' \in [0, 1]$, for $u, \delta \geq 0$ and $1 < P \leq 2$,

$$\frac{K_b(P(u + \delta)) - K_b(Pu)}{P} \leq 2u\delta + \delta^2. \quad (5.18)$$

Moreover, differentiating $K_b(Pu)/P$ and using (2.6) gives

$$0 \leq \frac{\partial}{\partial P} \frac{K_b(Pu)}{P} \leq u^2. \quad (5.19)$$

As

$$P - P^0 = PP^0 \left(\frac{1}{P^0} - \frac{1}{P} \right) = \frac{PP^0}{Q_1} \leq \frac{16R_i}{d}, \quad (5.20)$$

we obtain

$$\begin{aligned} & \frac{1}{P} \sum_j K_{b_j}(P(u_j^0 + \delta u_j)) - \frac{1}{P^0} \sum_j K_{b_j}(P^0 u_j^0) \\ & \leq 2 \sum_j u_j^0 \delta u_j + \sum_j (\delta u_j)^2 + \frac{16R_i}{d} \sum_j (u_j^0)^2. \end{aligned} \quad (5.21)$$

Each term is now bounded without probabilistic input. From (5.3), (5.9), $m_j \leq M_w$, and $k/\sqrt{d} \leq 1/D$,

$$2 \sum_j u_j^0 \delta u_j \leq \frac{4m_0 M_w^2 S_i}{D} \frac{S_i}{d}, \quad (5.22)$$

$$\sum_j (\delta u_j)^2 \leq \frac{2M_w^2 a_+ S_i}{D^2} \frac{S_i}{d}, \quad (5.23)$$

$$\frac{16R_i}{d} \sum_j (u_j^0)^2 \leq \frac{32m_0^2 M_w^2 a_+ S_i}{a_- D^2} \frac{S_i}{d}. \quad (5.24)$$

For completeness, (5.22) uses $\sum_j \delta u_j \leq 2M_w S_i/d$; (5.23) uses $\max_j \delta u_j \leq M_w R_i/d$; and (5.24) uses

$$\sum_j (u_j^0)^2 \leq \frac{2m_0^2 M_w^2 S_i}{a_-} \frac{S_i}{d}, \quad (5.25)$$

which follows from the first inequality in (5.3) and $r - 2 \geq k - 1$. The sum of (5.22)–(5.24) is precisely $\hat{L}_w(C) S_i/d$.

Finally, at the central cutoff the deterministic Δ -part of (5.7) is $-m_0^2 S_i/d$. Combining it with (5.15) and (5.21)–(5.24) leaves

$$-\hat{B}_w(C) \frac{S_i}{d}. \quad (5.26)$$

Equations (5.16)–(5.17) charge the old exact-CGF deficit once; (5.26) charges only the new increment factor and the change $P^0 \rightarrow P$; and (4.6) pays the boundary residual. Hence the three ledger entries are disjoint on the same conditional history.

6. REVERSE INDUCTION AND PERFECT-SEQUENCE EXTRACTION

By (7.7)–(7.11), after increasing the existential threshold C_0 and restricting w to a sufficiently small right-neighborhood of ω_0 , we have $\hat{B}_w(C) > 0$, $E_w(C) > 0$, and $\mu_w > 0$. We fix this range throughout the section; in

particular, every monotonicity assertion below has its sign justified before it is used.

Define the new accumulated deterministic deficit

$$\mathcal{N}_{C,w,d}(r) = \frac{1}{d} \left[2\hat{B}_w A_0 \binom{r}{4} + E_w K_r \right]. \quad (6.1)$$

At an exposure step with $k = r - i$ future vertices, the same-history calculation of Section 5, followed by the pathwise boundary estimate (4.6), proves the one-step kernel bound (2.37) with

$$\lambda_i = \mathbf{d}_{k,w,d} + \hat{B}_w(C) \frac{S_i}{d} - \sum_{j>i} \frac{e_{ij}^2}{2\mu_w d} - \frac{d^{-1/5}}{\sqrt{d}} \sum_{j>i} T_{ij}. \quad (6.2)$$

Every coefficient is deterministic and fixed before column i is sampled. Equations (2.33) and (2.40) retain every future coefficient T_{jq} , while (4.6) supplies each new boundary edge ij with its full coefficient T_{ij} . Thus the exact implication (S3) applies without an exchangeability assumption.

Apply (2.38) at $s = 0$, average over $\mathcal{F}_0$, and then use Lemma 5.1. Since $\mathcal{P}_0^T = 0$, this gives

$$P_{R,r}^* \leq \mathcal{H}_C(r) \exp[-\mathcal{D}_{C,w,d}(r) - \mathcal{N}_{C,w,d}(r) + J_{r,d}], \quad (6.3)$$

where

$$0 \leq J_{r,d} \leq \frac{m_0 + \theta/D}{1 - \rho} d^{-1/5} \binom{r}{2} = o_C(r^2). \quad (6.4)$$

Here $J_{r,d}$ denotes the displayed deterministic upper envelope, so $J_{q,d} \leq J_{\ell,d}$ for $q \leq \ell$. The base cases $r = 0, 1, 2$ follow from the empty-event convention and the same one-step identity; no asymptotic estimate is needed there.

6.1. Uniform restoration after deleting vertices. Let extraction retain $q = \ell - u$ vertices. Relabel them in retained order and rebuild the deterministic array for target size q . Since $q/\sqrt{d} \leq \ell/\sqrt{d}$, all row-sum, Hessian and Hölder feasibility inequalities only improve.

First, the bracket in (2.14) lies in $[0, 1]$, and $m_w \leq m_0$. Consequently

$$\begin{aligned} 0 &\leq \mathcal{D}_{C,w,d}(\ell) - \mathcal{D}_{C,w,d}(\ell - u) \\ &\leq \frac{m_0^4}{d} \sum_{k=\ell-u}^{\ell-1} k(k-1)^2 \leq \frac{m_0^4}{D^2} \ell u. \end{aligned} \quad (6.5)$$

For sufficiently large C , $m_0^4/D^2 = (1 + o(1))/(8 \log C) < 1$, so the right side is at most ℓu .

Next,

$$\binom{\ell}{4} - \binom{\ell-u}{4} \leq u \binom{\ell}{3}, \quad \binom{\ell}{3} - \binom{\ell-u}{3} \leq u \binom{\ell}{2}. \quad (6.6)$$

Using $K_r = \binom{r}{3} + 2\binom{r}{4}$, (6.6) yields

$$\begin{aligned} 0 &\leq \mathcal{N}_{C,w,d}(\ell) - \mathcal{N}_{C,w,d}(\ell - u) \\ &\leq \frac{1}{d} \left[2\hat{B}_w A_0 u \binom{\ell}{3} + E_w \left(u \binom{\ell}{2} + 2u \binom{\ell}{3} \right) \right] \\ &\leq \left[\frac{\hat{B}_w A_0 + E_w}{3D^2} + \frac{E_w}{2D^2\ell} \right] \ell u. \end{aligned} \quad (6.7)$$

The last bracket is $O(1/\log C) + O_C(\ell^{-1}) < 1$ for sufficiently large C and then sufficiently large ℓ , so (6.7) also costs at most ℓu .

Finally, the ratio between the source envelopes at sizes $\ell - u$ and ℓ costs at most

$$[\log(1/p) + 1] \ell u. \quad (6.8)$$

Indeed, $\binom{\ell}{2} - \binom{\ell-u}{2} \leq \ell u$ pays the missing p -factors, and $\binom{\ell}{3} - \binom{\ell-u}{3} \leq \ell^2 u/2$, together with $\vartheta_C a^3/(p^3 D) \leq 2$ for all sufficiently large C , pays the cubic term. The positive $d^{-1/5}\binom{r}{2}$ and r terms in $\mathcal{H}_C(r)$, as well as (6.4), are monotone in r and require no restoration charge.

6.2. The binomial subset sum. Apply the pinned extraction inequality (2.42), then (6.3) at $q = \ell - u$, and finally (6.5)–(6.8). For every specified retained set I and every $1 \leq u \leq \ell$,

$$\begin{aligned} p_I &\leq P_{R,\ell-u}^* \left(\frac{p}{10} \right)^{10\ell u} \\ &\leq \mathcal{H}_C(\ell) \exp[-\mathcal{D}_{C,w,d}(\ell) - \mathcal{N}_{C,w,d}(\ell) + J_{\ell,d}] e^{-\eta_C \ell u}, \end{aligned} \quad (6.9)$$

where the explicit residual exponent

$$\eta_C = 10 \log(10/p) - \log(1/p) - 3 > 0 \quad (6.10)$$

pays, respectively, the source-envelope, old-CGF, and new-residual restoration charges. Summing over all possible deleted sets is now an exact binomial calculation:

$$\sum_{u=1}^{\ell} \binom{\ell}{u} e^{-\eta_C \ell u} = (1 + e^{-\eta_C \ell})^{\ell} - 1 = o(1). \quad (6.11)$$

Including the $u = 0$ perfect term proves the unconditional red probability bound

$$P_{R,\ell} \leq \mathcal{H}_C(\ell) \exp[-\mathcal{D}_{C,w,d}(\ell) - \mathcal{N}_{C,w,d}(\ell) + J_{\ell,d}] (1 + o(1)). \quad (6.12)$$

This proves extraction of the complete new exponent, rather than merely asserting that a source failure factor is large enough.

7. FIRST MOMENTS, RED-BLUE CROSSING AND ASYMPTOTICS

7.1. The two explicit first-moment thresholds. Fix C and $w > \omega_0$, and let $\ell \rightarrow \infty$. From (2.12), (2.18), (6.1), and (6.12),

$$\limsup_{\ell \rightarrow \infty} \frac{1}{\ell^2} \log P_{R,\ell} \leq \frac{1}{2} \log p - \vartheta_C \frac{a^3}{6p^3 D} - G(C, w) - \widehat{H}(C, w). \quad (7.1)$$

The discretization term vanishes because $J_{\ell,d} = o_C(\ell^2)$. Let $n = \lfloor e^{\sigma \ell} \rfloor$. Since $\binom{n}{\ell} \leq n^\ell$, the expected number of red K_ℓ 's tends to zero whenever

$$\begin{aligned} \sigma &< -\frac{1}{2} \log p + \vartheta_C \frac{a^3}{6p^3 D} + G(C, w) + \widehat{H}(C, w) \\ &= -\frac{1}{2} \log p_C + \frac{B_R(C)}{2} + G(C, w) + \widehat{H}(C, w). \end{aligned} \quad (7.2)$$

This proves the red threshold directly from the probability ledger.

For $q_\ell = \lfloor C\ell \rfloor$, (2.43) and $q_\ell/\ell = C + O(\ell^{-1})$ give

$$\limsup_{\ell \rightarrow \infty} \frac{1}{\ell^2} \log P_{B,q_\ell} \leq \frac{C^2}{2} \log(1-p) + \frac{2a^3 C^3}{3(1-p)^3 D}. \quad (7.3)$$

The bound $\binom{n}{q_\ell} \leq n^{q_\ell}$ therefore makes the expected number of blue K_{q_ℓ} 's tend to zero whenever

$$\sigma < -\frac{C}{2} \log(1-p) - \frac{2a^3 C^2}{3(1-p)^3 D} = -\frac{1}{2} \log p_C + \frac{CB_B(C)}{2}. \quad (7.4)$$

No blue improvement is used. A union bound over red and blue vertex sets therefore produces a coloring with neither forbidden clique for every σ strictly below the minimum of the right sides of (7.2) and (7.4).

7.2. The crossing is separated by a constant. The pinned Gaussian asymptotics, all in the order $C \rightarrow \infty$, are

$$\begin{aligned} \frac{p}{p_C} &= 1 + O((\log C)^{-1}), & \frac{a^3}{p^3 D} &= \frac{1}{2} + o(1), & \vartheta_C &= 1 - o(1), \\ B_R(C) &= \frac{1}{6} + o(1), & CB_B(C) &= 1 + o(1). \end{aligned} \quad (7.5)$$

The exact integral (2.17) also gives

$$G_*(C) = \frac{1 + o(1)}{32 \log C} = o(1). \quad (7.6)$$

Indeed $m(c - \omega_0)/m_0 \rightarrow 1$, $P_*(x) \rightarrow 1$ uniformly, and the bracket in (2.17) tends uniformly to one, while $m_0^4/D^2 = (1 + o(1))/(8 \log C)$.

The controlled-residual constants satisfy

$$\begin{aligned} \rho &= o(1), & m'_0 &\rightarrow 1, & m_0 m''(c) &= o(1), \\ M_w/m_0 &\rightarrow 1, & j_\pm(w) &\rightarrow 1, & \mu_w &\rightarrow 1, \end{aligned} \quad (7.7)$$

where the right limit $w \downarrow \omega_0$ is taken for each fixed C before $C \rightarrow \infty$, and

$$\hat{L}_w = o(m_0^2), \quad \hat{B}_w = m_0^2[1 + o(1)]. \quad (7.8)$$

Thus every positivity and feasibility condition used above holds for all sufficiently large fixed C , uniformly for w in a sufficiently small right-neighborhood of ω_0 .

Equations (7.5)–(7.8) show

$$\frac{CB_B(C)}{2} - \left[\frac{B_R(C)}{2} + G_*(C) + \hat{H}_*(C) \right] = \frac{5}{12} + o(1) > 0. \quad (7.9)$$

This positive limiting gap, rather than the mere fact that the added terms are $o(1)$, proves that red remains the bottleneck. Taking the supremum of (7.2) over $w > \omega_0$, using continuity as $w \downarrow \omega_0$, and then using (7.9) proves Theorem 1.1.

7.3. The new leading coefficient. Finally,

$$m_0^2 = (2 + o(1)) \log C, \quad D^2 = (32 + o(1))(\log C)^3, \quad (7.10)$$

and hence

$$\hat{B}_w A_0 = (4 + o(1))(\log C)^2, \quad E_w = (2 + o(1))(\log C)^2. \quad (7.11)$$

Substitution into (2.23) gives

$$\hat{H}_*(C) = \frac{(6 + o(1))(\log C)^2}{12(32 + o(1))(\log C)^3} = \frac{1 + o(1)}{64 \log C}. \quad (7.12)$$

The exact-merge predecessor is $H_*(C) = (1 + o(1))/(96 \log C)$, so the new term is eventually strictly larger. Since $B_R(C)/2 \rightarrow 1/12$, the public coarsening $B_R(C)/2 \geq 1/20$ also follows after increasing the existential threshold.

7.4. A precisely delimited scalar method cap.

Definition 7.1 (Admissible scalar residual family). Fix constants $M_0, M_{\text{ext}} > 0$. An (M_0, M_{ext}) -admissible family $\mathcal{E}$ assigns, for every sufficiently large C , every w in a right-neighborhood of ω_0 , and every $0 \leq r \leq \ell$, deterministic data

$$(e^{C,w,\ell,r}, T^{C,w,\ell,r}, \beta_{C,w,\ell,r}, \mu_{C,w,\ell,r}).$$

The same constants M_0, M_{ext} apply to every member of the family and to every C, w, ℓ, r . At $r = 0, 1$, the arrays are empty and we set $\beta = 0, \mu = 1$, and $\mathcal{G}_{\mathcal{E}} = 0$.

(A1) The increments are fixed before exposure and

$$0 \leq e_{ij} \leq M_0 m_0^2 r \quad (1 \leq i < j \leq r). \quad (7.13)$$

(A2) The array is the unique solution of the generalized recursion

$$T_{ij} = t_0 + \gamma_d \sum_{\substack{q > i \\ q \neq j}} T_{jq} + e_{ij}, \quad \Delta_{ij} = T_{ij} - t_0 \geq 0. \quad (7.14)$$

Thus neither the recursion nor the admissible class refers to the particular choice $e_{ij} = \theta(i - 1)$.

(A3) Put

$$S_i = \sum_{\{j,q\} \subset V_i} \Delta_{jq}, \quad R_i = \max_{j \in V_i} \sum_{q \neq j} \Delta_{jq}, \quad k = |V_i|.$$

The three-factor split is feasible at every step:

$$\frac{4t_0k}{d} + \frac{4R_i}{d} \leq \frac{1}{2}. \quad (7.15)$$

For $r \geq 2$, writing F_i^T for (3.1) with coefficient array T , the two scalar coefficients obey the exact constraints

$$0 < \mu_{C,w,\ell,r} \leq \inf_{1 \leq i < r} \inf_{b \in [c-w, c+w]^{V_i}} \lambda_{\min}(-\nabla^2 F_i^T(b)), \quad 0 \leq \beta_{C,w,\ell,r} \leq m_0^2. \quad (7.16)$$

With these coefficients, define

$$\lambda_i^\mathcal{E} = \mathbf{d}_{r-i,w,d} + \beta_{C,w,\ell,r} \frac{S_i}{d} - \sum_{j>i} \frac{e_{ij}^2}{2\mu_{C,w,\ell,r}d} - \frac{d^{-1/5}}{\sqrt{d}} \sum_{j>i} T_{ij}. \quad (7.17)$$

The family must certify, on every history in $\mathfrak{H}_{r,i-1}^R$, the one-step inequality

$$\mathcal{K}_i^T \leq \frac{\mathcal{H}_C(r-i+1)}{\mathcal{H}_C(r-i)} e^{-\lambda_i^\mathcal{E}}. \quad (7.18)$$

Thus this scalar class credits exactly the displayed $\beta S_i/d$ term, pays the displayed scalar square penalty, and discards every favorable term not listed in (7.17).

(A4) Relative to the exact weighted rule (S3), the family's credited deterministic contribution is the normalized functional

$$\mathcal{G}_\mathcal{E}(C, w, \ell, r) = \frac{1}{d} \left[\beta_{C,w,\ell,r} \sum_{i=1}^{r-1} S_i - \frac{1}{2\mu_{C,w,\ell,r}} \sum_{1 \leq i < j \leq r} e_{ij}^2 \right]. \quad (7.19)$$

This definition excludes adaptive weights, matrix-valued completion, and unlisted source credits.

(A5) After deleting u vertices, relabeling in retained order, and rebuilding the family member, with $x_+ = \max\{x, 0\}$,

$$0 \leq [\mathcal{G}_\mathcal{E}(C, w, \ell, \ell) - \mathcal{G}_\mathcal{E}(C, w, \ell, \ell - u)]_+ \leq \frac{M_{\text{ext}}}{\log C} \ell u \quad (7.20)$$

for every $0 \leq u \leq \ell$. Thus extraction is a quantified condition, not a reference to the optimizer-specific estimate.

Define

$$\begin{aligned} \text{Lead}_C(\mathcal{E}) &= \limsup_{w \downarrow \omega_0} \limsup_{\ell \rightarrow \infty} \frac{\mathcal{G}_\mathcal{E}(C, w, \ell, \ell)}{\ell^2}, \\ \mathfrak{M}_{M_0, M_{\text{ext}}}(C) &= \sup_{\mathcal{E}} \text{Lead}_C(\mathcal{E}), \end{aligned} \quad (7.21)$$

where the supremum is over (M_0, M_{ext}) -admissible families. The right-neighborhood in the first limsup may depend on C , but not on ℓ or r .

Proposition 7.2 (Cap within Definition 7.1). *For every fixed M_0, M_{ext} , the leading functional in (7.21) satisfies*

$$\limsup_{C \rightarrow \infty} 64 \log C \mathfrak{M}_{M_0, M_{\text{ext}}}(C) \leq 1. \quad (7.22)$$

There are fixed absolute constants M_0^, M_{ext}^* for which the family $e_{ij} = \theta(i-1)$, $\beta = \hat{B}_w(C)$, and $\mu = \mu_w$ is admissible and satisfies*

$$\lim_{C \rightarrow \infty} 64 \log C \text{Lead}_C(\mathcal{E}^*) = 1. \quad (7.23)$$

Thus $1/64$ is both the cap and an attained leading coefficient within the precisely delimited scalar class, for fixed admissibility constants large enough to contain $\mathcal{E}^$.*

Proof. Fix an admissible family and abbreviate its two scalar coefficients by β and μ . The exact constraint (7.16) gives

$$\beta \leq m_0^2. \quad (7.24)$$

At $b = c\mathbf{1}$, the row condition (7.15) and the diagonal formula (A.1) give, for every $j \in V_i$,

$$\mu \leq \lambda_{\min}(-\nabla^2 F_i^T(c\mathbf{1})) \leq (-\nabla^2 F_i^T(c\mathbf{1}))_{jj} \leq m'_0 + \frac{1}{8}m_0 m''(c) = 1 + o(1). \quad (7.25)$$

Here $t_0 k + R_i \leq d/8$, $m'_0 \rightarrow 1$, and $m_0 m''(c) = o(1)$. Hence every valid common scalar completion constant satisfies, uniformly over the class,

$$\mu \leq 1 + o(1). \quad (7.26)$$

Let $\mathcal{L}$ be the feedback operator in (7.14). Its row norm is at most $\gamma_d r \leq m_0 m'_0 / D = \rho = o(1)$. Writing

$$g_{ij} = A_0(r - i - 1),$$

the recursion for Δ is $\Delta = g + \mathcal{L}\Delta + e$. Therefore

$$\Delta = (I - \mathcal{L})^{-1}g + e + O(\rho m_0^2 r) \quad (7.27)$$

uniformly on every edge, with an implicit constant depending only on M_0 , by (7.13). The exact exposure multiplicity is

$$\sum_{i=1}^{r-1} S_i = \sum_{1 \leq j < q \leq r} (j-1) \Delta_{jq}. \quad (7.28)$$

The first term of (7.27) contributes

$$2A_0 \binom{r}{4} [1 + O(\rho)] \quad (7.29)$$

to (7.28). The last term changes the numerator in (7.19) by at most $O_{M_0}(\rho m_0^4 r^4) = o(m_0^4 r^4)$.

For the direct residual term, scalar square completion is pointwise:

$$\beta(j-1)e_{jq} - \frac{e_{jq}^2}{2\mu} \leq \frac{\mu\beta^2}{2}(j-1)^2. \quad (7.30)$$

Summing (7.30) gives $\mu\beta^2 K_r/2$. Set $r = \ell$, divide (7.19) by ℓ^2 , and use $\binom{\ell}{4}/\ell^4 \rightarrow 1/24$, $K_\ell/\ell^4 \rightarrow 1/12$, and $d/\ell^2 \rightarrow D^2$. Equations (7.24)–(7.30) yield

$$\text{Lead}_C(\mathcal{E}) \leq \frac{\beta A_0}{12D^2} + \frac{\mu\beta^2}{24D^2} + o\left(\frac{m_0^4}{D^2}\right) \leq \frac{m_0^4}{8D^2}[1 + o(1)] = \frac{1 + o(1)}{64 \log C}. \quad (7.31)$$

The error is uniform for fixed M_0, M_{ext} , so taking the supremum and then the limsup proves (7.22).

For $e_{ij} = \theta(i-1)$, equations (5.5), (7.7), and (7.8) make (7.30) asymptotically tight. Equations (5.11), (6.2), and (4.4) verify (7.15)–(7.18), and (7.13) holds with $M_0^* = 1$. To verify (7.20), write $\Delta^{(r)}$ for the array rebuilt at target r , and put $r_- = \ell - u$. Subtracting the two positive recursions (7.14), and using $\|\mathcal{L}\|_\infty \leq \rho < 1/2$, gives absolute constants K_1, K_2 such that

$$\max_{1 \leq j < s \leq r_-} \left| \Delta_{js}^{(\ell)} - \Delta_{js}^{(r_-)} \right| \leq K_1 m_0^2 u, \quad \max_{j < s \leq \ell} \Delta_{js}^{(\ell)} \leq K_2 m_0^2 \ell. \quad (7.32)$$

Indeed, the second bound is (5.1). If U denotes the first maximum, subtraction of (7.14) gives $U \leq A_0 u + \rho U + \gamma_d u K_2 m_0^2 \ell$. Now $A_0 = O(m_0^2)$, $\gamma_d \ell \leq \rho$, and $\rho < 1/2$ give the first bound. The multiplicity identity (7.28) and (5.4) now give

$$\left| \sum_i S_i^{(\ell)} - \sum_i S_i^{(r_-)} \right| \leq K_3 m_0^2 \ell^3 u, \quad |K_\ell - K_{r_-}| \leq K_4 \ell^3 u \quad (7.33)$$

for absolute K_3, K_4 . Since $\beta \leq m_0^2$, $\mu_w \rightarrow 1$, and $m_0^4/D^2 = O(1/\log C)$, (7.33) proves (7.20) for some absolute M_{ext}^* . Finally, the functional is exactly (2.23) up to $o(1/\log C)$, and (7.12) gives the asserted limit (7.23). $\square$

The proposition does not quantify over outcome-adaptive weights, matrix-valued square completion, higher-order potentials, a sharper source quadratic MGF, or a different extraction invariant.

8. REPRODUCIBILITY AND CLAIM BOUNDARY

The proof specification, arithmetic checker and independent source-level review have SHA-256 prefixes **2234a769**, **616040b9** and **ef6ddbce**, respectively. The checker replays the recursion, multiplicity identities, Hessian samples, exact-CGF increments, finite margins at four large C -values and convergence to the leading constant. It is an arithmetic witness, not a substitute for the reverse-induction proof.

From the repository root run

```
bash scripts/bootstrap.sh
bash scripts/reproduce_lower.sh
```

to reproduce the bridge, exact-CGF predecessor, nonexchangeable ledger and controlled-residual calculations with pinned dependencies.

The theorem is source-relative, fixed- C , asymptotic in ℓ , and non-effective in C_0 . It supplies no new blue-side construction and no finite Ramsey number. The current manuscript does not claim a common- normalization strict improvement over the final printed Lin–Niu theorem, global optimality, priority, publication, or world-best status.

Logical proof status. As a standalone theorem without Source interface 2.2, (1.1) is PROVABLE AFTER WEAKENING / EXTRA ASSUMPTION, not an unconditional claim. Under the explicit items (S1)–(S5), the weighted one-step comparison, residual contribution, extraction algebra, and crossing argument are PROVABLE AS STATED. In particular, the manuscript does not replace the weighted source interface by numerical evidence.

APPENDIX A. DETAILED ONE-STEP LEDGER ESTIMATES

This appendix records the estimates used in Sections 4 and 5 in a form independent of the exploratory calculation log. Throughout, C is fixed and sufficiently large, $w > \omega_0$, and the target set exposed after column i is $V_i = \{i + 1, \dots, r\}$, of size $k = r - i$.

A.1. The Hessian on the entire cutoff box. Recall

$$F_i(b) = \sum_{j \in V_i} \log \Phi(-b_j) - \frac{1}{d} \sum_{\{j,q\} \subset V_i} T_{jq} m(b_j) m(b_q).$$

Since $(\log \Phi(-x))' = -m(x)$, the entries of the Hessian are

$$\partial_{jj}^2 F_i(b) = -m'(b_j) - \frac{m''(b_j)}{d} \sum_{q \in V_i \setminus \{j\}} T_{jq} m(b_q), \quad (\text{A.1})$$

$$\partial_{jq}^2 F_i(b) = -\frac{T_{jq}}{d} m'(b_j) m'(b_q), \quad j \neq q. \quad (\text{A.2})$$

The inverse Mills ratio satisfies $m'' \geq 0$. Hence the interaction term in (A.1) can only strengthen the diagonal of $-\nabla^2 F_i$. On $[c - w, c + w]^{V_i}$, monotonicity gives

$$m'(b_j) \geq j_-(w), \quad |m'(b_j) m'(b_q)| \leq j_+(w)^2. \quad (\text{A.3})$$

The residual recursion and $r/\sqrt{d} \leq 1/D$ imply

$$T_{\max} \leq \frac{t_0 + \theta r}{1 - \rho}, \quad (\text{A.4})$$

$$\frac{1}{d} \max_j \sum_{q \neq j} T_{jq} \leq \frac{m_0 + \theta/D}{D(1 - \rho)}. \quad (\text{A.5})$$

Combining (A.1)–(A.5) with Gershgorin's theorem proves

$$-\nabla^2 F_i(b) \succeq \left[j_-(w) - j_+(w)^2 \frac{m_0 + \theta/D}{D(1 - \rho)} \right] I = \mu_w I. \quad (\text{A.6})$$

This is a full-box statement, not an expansion valid only at the central cutoff.

A.2. Boundary merge with a controlled residual. The gradient at $c\mathbf{1}$ is

$$\partial_j F_i(c\mathbf{1}) = -m_0 - \frac{m_0 m'_0}{d} \sum_{q \in V_i \setminus \{j\}} T_{jq}. \quad (\text{A.7})$$

The defining recursion

$$T_{ij} = m_0 \sqrt{d} + \frac{m_0 m'_0}{\sqrt{d}} \sum_{q \in V_i \setminus \{j\}} T_{jq} + e_{ij} \quad (\text{A.8})$$

therefore gives

$$\partial_j F_i(c\mathbf{1}) = -\frac{T_{ij} - e_{ij}}{\sqrt{d}}. \quad (\text{A.9})$$

For $\delta_j = b_j - c = \sqrt{d} h_{ij} + \varepsilon_{ij}$, with $|\varepsilon_{ij}| \leq d^{-1/5}$, the tangent inequality from (A.6) gives

$$\begin{aligned} F_i(b) &\leq F_i(c\mathbf{1}) - \sum_{j \in V_i} T_{ij} h_{ij} + \sum_{j \in V_i} \left(\frac{e_{ij} \delta_j}{\sqrt{d}} - \frac{\mu_w \delta_j^2}{2} \right) + \frac{d^{-1/5}}{\sqrt{d}} \sum_{j \in V_i} T_{ij} \\ &\leq F_i(c\mathbf{1}) - \sum_{j \in V_i} T_{ij} h_{ij} + \sum_{j \in V_i} \frac{e_{ij}^2}{2\mu_w d} + \frac{d^{-1/5}}{\sqrt{d}} \sum_{j \in V_i} T_{ij}. \end{aligned} \quad (\text{A.10})$$

The second line is the scalar square completion $xy - \mu_w y^2/2 \leq x^2/(2\mu_w)$. Thus every boundary edge merges with its full coefficient T_{ij} ; only the explicit residual is paid.

A.3. Mass identities. Let $\Delta_{ij} = T_{ij} - t_0$. Backward induction in the first endpoint gives, for every $j < q$,

$$A_0(r - j - 1) + \theta(j - 1) \leq \Delta_{jq} \leq \frac{A_0 + \rho\theta}{1 - \rho}(r - j - 1) + \theta(j - 1). \quad (\text{A.11})$$

For S_i and R_i as in Section 5, this implies

$$S_i \geq a_-(r - 2) \binom{k}{2}, \quad R_i \leq a_+(r - 2)(k - 1), \quad \frac{R_i}{S_i} \leq \frac{2a_+}{a_-k}. \quad (\text{A.12})$$

The exact accumulated sums are obtained by exchanging the order of summation. The future-degree part of (A.11) counts each ordered quadruple twice, while the residual part contributes

$$K_r := \sum_{i=1}^{r-1} (r-i)(i-1)^2 = \binom{r}{3} + 2\binom{r}{4}. \quad (\text{A.13})$$

Consequently

$$\sum_{i=1}^{r-1} S_i \geq 2A_0 \binom{r}{4} + \theta K_r, \quad \sum_{1 \leq i < j \leq r} e_{ij}^2 = \theta^2 K_r. \quad (\text{A.14})$$

No asymptotic replacement is used in (A.13)–(A.14).

A.4. Three-factor Hölder comparison. Condition on A_i^R and $\mathcal{F}_{i-1}$, on a history in B_r^R , and write

$$X_j = -\frac{m(b_j)}{\sqrt{d}} + \frac{\zeta_j}{\sqrt{d}}, \quad u_j^0 = \frac{t_0}{d} \sum_{q \neq j} m(b_q), \quad \delta u_j = \frac{1}{d} \sum_{q \neq j} \Delta_{jq} m(b_q). \quad (\text{A.15})$$

The variables ζ_j are conditionally independent, centered and 1-subgaussian. The centered exponent is exactly

$$\sum_j (u_j^0 + \delta u_j) \zeta_j - \frac{t_0}{d} \sum_{j < q} \zeta_j \zeta_q - \frac{1}{d} \sum_{j < q} \Delta_{jq} \zeta_j \zeta_q. \quad (\text{A.16})$$

Use conjugate exponents

$$Q_0 = \frac{d}{4t_0k}, \quad Q_1 = \frac{d}{4R_i}, \quad P^{-1} = 1 - Q_0^{-1} - Q_1^{-1}. \quad (\text{A.17})$$

They are all greater than one, with $P \leq 2$, because

$$\frac{4m_0}{D} + \frac{4a_+}{D^2} < \frac{1}{2}. \quad (\text{A.18})$$

Hölder gives one linear-CGF factor and two quadratic factors. The uniform t_0 factor is the frozen HMS factor. For the increment, use

$$-\sum_{j < q} \Delta_{jq} \zeta_j \zeta_q \leq \frac{1}{2} \sum_j R_j \zeta_j^2, \quad \frac{Q_1 R_j}{2d} \leq \frac{1}{8}. \quad (\text{A.19})$$

The square-MGF estimate $\log \mathbb{E} e^{x \zeta_j^2} \leq (16/3)x$ for $0 \leq x \leq 1/8$ therefore gives

$$\frac{1}{Q_1} \sum_j \frac{16}{3} \frac{Q_1 R_j}{2d} = \frac{16}{3} \frac{S_i}{d}. \quad (\text{A.20})$$

For the linear factor, let $P^0 = P_{k,d}^0$. The source exact-CGF deficit on this identical history is

$$\mathfrak{d}_i(b) = \sum_j \left[(u_j^0)^2 - \frac{1}{P^0} K_{b_j}(P^0 u_j^0) \right] \geq \mathfrak{d}_{k,w,d}. \quad (\text{A.21})$$

Thus it remains only to bound the change from (P^0, u^0) to $(P, u^0 + \delta u)$. Since $0 \leq K_b'' \leq 1$,

$$\frac{K_b(P(u + \delta)) - K_b(Pu)}{P} \leq 2u\delta + \delta^2, \quad (\text{A.22})$$

$$0 \leq \frac{\partial}{\partial P} \frac{K_b(Pu)}{P} \leq u^2. \quad (\text{A.23})$$

Furthermore

$$P - P^0 \leq \frac{16R_i}{d}, \quad \sum_j \delta u_j \leq \frac{2M_w S_i}{d}, \quad \max_j \delta u_j \leq \frac{M_w R_i}{d}, \quad (\text{A.24})$$

and

$$\sum_j (u_j^0)^2 \leq \frac{2m_0^2 M_w^2 S_i}{a_- d}. \quad (\text{A.25})$$

Substitution in (A.22)–(A.24) gives, in order,

$$\frac{4m_0 M_w^2 S_i}{D} \frac{1}{d}, \quad \frac{2M_w^2 a_+ S_i}{D^2} \frac{1}{d}, \quad \frac{32m_0^2 M_w^2 a_+ S_i}{a_- D^2} \frac{1}{d}. \quad (\text{A.26})$$

These are precisely the three summands of $\hat{L}_w(C)S_i/d$. Combining them with (A.20) and the central deterministic gain $-m_0^2 S_i/d$ gives

$$-\left(m_0^2 - \frac{16}{3} - \hat{L}_w(C)\right) \frac{S_i}{d} = -\hat{B}_w(C) \frac{S_i}{d}. \quad (\text{A.27})$$

Equation (A.21) is charged once, (A.27) pays only the new $\Delta = T - t_0$ factor and the change of exponent, and (A.10) pays the residual gradient. This is the promised same-history separation.

A.5. One-step conclusion and extraction. Combining (A.10), (A.21), and (A.27) proves (2.37) with the value λ_i in (6.2). The exact weighted reverse-induction rule (S3), followed by (A.14), then gives the empty-state bound

$$P_{R,r}^* \leq \mathcal{H}_C(r) \exp \left[-\mathcal{D}_{C,w,d}(r) - \frac{1}{d} \left\{ 2\hat{B}_w A_0 \binom{r}{4} + E_w K_r \right\} + J_{r,d} \right]. \quad (\text{A.28})$$

where

$$0 \leq J_{r,d} \leq \frac{m_0 + \theta/D}{1 - \rho} d^{-1/5} \binom{r}{2} = o_C(r^2). \quad (\text{A.29})$$

If extraction deletes u vertices from a retained set of size ℓ , then the old exact-CGF restoration cost is at most

$$\mathcal{D}_{C,w,d}(\ell) - \mathcal{D}_{C,w,d}(\ell - u) \leq \frac{m_0^4}{D^2} \ell u \leq \ell u, \quad (\text{A.30})$$

and the loss in the new deterministic exponent is at most

$$\frac{1}{d} \left[2\hat{B}_w A_0 u \binom{\ell}{3} + E_w \left(u \binom{\ell}{2} + 2u \binom{\ell}{3} \right) \right] \leq \left[\frac{\hat{B}_w A_0 + E_w}{3D^2} + \frac{E_w}{2D^2 \ell} \right] \ell u. \quad (\text{A.31})$$

For sufficiently large C and then sufficiently large ℓ , the bracket in (A.31) is

$$O(1/\log C) + O_C(\ell^{-1}) < 1.$$

Together with the source-envelope restoration $[\log(1/p) + 1]\ell u$, the total cost is at most $[\log(1/p) + 3]\ell u$. The pinned failure factor therefore leaves $\eta_C = 10\log(10/p) - \log(1/p) - 3 > 0$, and

$$\sum_{u=1}^{\ell} \binom{\ell}{u} e^{-\eta_C \ell u} = (1 + e^{-\eta_C \ell})^{\ell} - 1 = o(1). \quad (\text{A.32})$$

This proves that (A.28) survives perfect-sequence extraction with its entire old and new exponent intact.

APPENDIX B. PINNED SOURCE INTERFACE AND QUANTIFIER LEDGER

This appendix records the exact external boundary of the lower-bound argument. It separates version-pinned conditional probability statements from the explicit weighted-extension rule and imports neither the first-moment conversion nor the Ramsey conclusion proved in Section 7.

B.1. Content-addressed source snapshots. The source files are Hunter–Milojević–Sudakov, arXiv:2512.17718v2, file `arXiv_version.tex`, SHA-256

b72958ac35554eccb94dedab5800349d2c021af7d60f767125cb46998e0fd54a

and Lin–Niu, arXiv:2605.25843v2, file `off-diagonal-Ramsey-R.tex`, SHA-256

dbf1bfa2c7603c81b2e56a97028b138ccf9a9299a3d1f20c0b08cf489136f429

The repository manifest records the versioned retrieval URLs and independently checks these hashes.

B.2. Source-line map for the source interface.

Pinned source lines	Exact use in this paper
HMS 145–170, 443–498	The Gaussian graph, Bartlett representation, column exposure, edge events, perfect events, and the unweighted projection state. These are the probability-space and history definitions in (2.24)–(2.33).
HMS 955–971, 1192–1218	Optimized red-perfectness and the exact cutoff approximation. Cauchy–Schwarz gives $\sqrt{d} h_{ij} \leq \alpha_R^2 \ell / \sqrt{d} \leq \omega_0$, while the diagonal approximation contributes $d^{-1/5}$. This is (S1).
HMS 330–354, 878–935	The subgaussian definition, square-MGF lemma, pre-Cauchy centered-linear allocation, and centered quadratic estimate. These give the quantified statements (2.35)–(2.36), the uniform t_0 factor in (5.12), and the square estimate in (5.14); no final Cauchy coarsening is imported.
Lin–Niu 359–493, 499–537	Monotonicity of upper-truncated Gaussian variance and the exact positive-direction CGF. After replacing their upper cutoff by $-b$, these give (2.4)–(2.6) and Lemma 2.1.
HMS 1192–1289	The red law of total probability, cutoff conditioning, projection merge, and scalar reverse-induction allocation. Equations (2.33), (2.39), and (2.40) write their weighted algebraic extension explicitly; (2.41) is the frozen scalar baseline. The nonuniform estimate is proved locally and is not attributed verbatim to HMS.

Pinned source lines	Exact use in this paper
HMS 385–410 and 955–971	For every specified retained set, the red-perfectness failure factor is $(p/10)^{10\ell u}$. This is the sole external extraction input (2.42); the restoration estimates and binomial sum are proved in (6.5)–(6.11).
HMS 977–1181, 1294–1312, together with 385–410	The unchanged blue reverse induction, its perfect bound, and the same greedy extraction argument, yielding the raw probability inequality (2.43). The first-moment threshold (7.4) is derived locally.

The separately fetched Yang–Mao v1 member has SHA-256

155b7104ec5b6935a576ae9f2b161976a966b0b46bd2b69153c0934ca688da2a

agreeing with the repository manifest. It is not an input to the lower-bound component, so no Yang–Mao line range is invoked here; assigning one would misstate the lower component’s dependency boundary.

B.3. Why the cutoff window is ω_0 . On the event B_r^R ,

$$|b_{ij} - c| \leq \sqrt{d}|h_{ij}| + d^{-1/5} \leq \frac{\alpha_R^2 \ell}{\sqrt{d}} + d^{-1/5} \leq \omega_0 + d^{-1/5}. \quad (\text{B.1})$$

For every fixed $w > \omega_0$, choosing $\ell \geq \ell_0(C, w)$ makes the last expression at most w . The proof is first carried out for this open range of w , and only after the fixed- C , $\ell \rightarrow \infty$ limit is taken do we let $w \downarrow \omega_0$. No finite- ℓ history is claimed to lie in the closed boundary window. This also explains why p , rather than p_C , occurs in α_R and ω_0 .

B.4. The old exact-CGF entry is an input formula, not a conclusion. At a step with k future vertices, define

$$b_j = b_{ij}, \quad M_j = \sum_{q \neq j} m(b_q), \quad u_j^0 = \frac{m_0 M_j}{\sqrt{d}}. \quad (\text{B.2})$$

Stopping the HMS centered-moment calculation before its final Cauchy coarsening gives the allocation $\sum_j (u_j^0)^2$. Applying the exact positive-direction CGF with $P_{k,d}^0$ gives

$$\sum_j \frac{1}{P_{k,d}^0} K_{b_j}(P_{k,d}^0 u_j^0). \quad (\text{B.3})$$

Their difference is the same-history quantity $\mathfrak{d}_i(b)$ in (5.16). The cutoff monotonicities give (5.17), and summing its deterministic lower bound gives (2.15). With $k/\ell \rightarrow x$ and $\sqrt{d}/\ell \rightarrow D$, the corresponding Riemann sum is (2.17). Hence $G_*(C)$ is defined and charged before the controlled residual is added; it is not imported through a final Ramsey theorem.

B.5. Raw ledgers and the local crossing calculation. The red interface before the new two deficits is the explicit envelope

$$\mathcal{H}_C(r) = p^{\binom{r}{2}} \exp \left[-\vartheta_C \frac{a^3}{p^3 \sqrt{d}} \binom{r}{3} + \frac{a}{p} d^{-1/5} \binom{r}{2} + r \right]. \quad (\text{B.4})$$

The blue interface is the explicit probability estimate (2.43). Dividing their logarithms by ℓ^2 , and only then applying $\binom{n}{k} \leq n^k$, gives the red and blue thresholds (7.2) and (7.4). The comparison is not assumed: the source asymptotics imply

$$\frac{B_R(C)}{2} = \frac{1}{12} + o(1), \quad \frac{CB_B(C)}{2} = \frac{1}{2} + o(1), \quad (\text{B.5})$$

whereas

$$G_*(C) + \hat{H}_*(C) = O(1/\log C) = o(1). \quad (\text{B.6})$$

Thus the blue-minus-red gap is $5/12 + o(1)$, proving locally that red is the active constraint for sufficiently large fixed C .

B.6. Order of quantifiers and proof status. The absolute constant K in (2.10) is fixed before selecting the large- C threshold, but its numerical value is not provided by the pinned source. The theorem therefore has the order

$$\exists C_0 \forall C \geq C_0 \exists \ell_0(C) \forall \ell \geq \ell_0(C), \quad (\text{B.7})$$

followed by $\ell \rightarrow \infty$. Only after that fixed- C theorem is established is $C \rightarrow \infty$ used to obtain the coefficient $1/64$. No explicit value of C_0 or $\ell_0(C)$ is asserted.

The resulting theorem is source-relative: its local residual algebra, three-factor bridge, reverse-induction charge, extraction sum, first moments and red–blue crossing are proved here. Items (S1), (S2), (S4), and (S5) are version-pinned source hypotheses; (S3) is the separately stated weighted extension rule and is not attributed verbatim to either source. The arithmetic checker corroborates the finite formulas but does not replace these probabilistic interfaces.

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